Qizhou Deng

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Portfolio: qizhoud.github.io/qizhoudeng.github.io · GitHub: github.com/QizhouD

RESEARCH INTERESTS

Game AI and Computational Media: motion and trajectory generation for interactive digital humans; multimodal player modeling and performance analytics in games; search-based Game AI and procedural content generation.

EDUCATION BACKGROUND

Queen Mary University of London 09/2022 – 09/2023
M.Sc. in Computer Games

University of Liverpool 09/2020 – 07/2022
B.S. in Computer Science

Xi'an Jiaotong-Liverpool University 09/2018 – 07/2020
B.S. in Information and Computing Science

RESEARCH AND SELECTED PROJECTS

Language-conditioned generative trajectory planning for real-time digital humans 2026 – Present

Repo: github.com/QizhouD/language-conditioned-generative-trajectory-planning-for-real-time-digital-humans

- Research MVP for short-horizon, language-conditioned 2D root-trajectory planning for interactive digital humans: five intents (idle, walk, turn, stop, wave), candidates sampled at **10 Hz**, collision-aware rejection and ranking, and rolling replanning.
- Implemented a **conditional flow-matching** model and a **deterministic Transformer baseline** on synthetic navigation scenes; the baseline detours around obstacles with a collision loss and produces straighter empty-scene paths with path supervision (figures on the portfolio site and in the repo).
- Built the Unity embodiment layer (github.com/QizhouD/LLM-NPC): a language model tags each reply with one of the five intents, which becomes a plan request that drives the character's root at 10 Hz through the planner's trajectory contract; plans currently come from a procedural mock source, not the trained model.
- Next: stabilizing flow-matching trajectories, a learned plan source for the Unity embodiment, and full-body motion.

AI-FPS-Coach: offline CS2 practice review 2026 – Present

Repo: github.com/QizhouD/AI-FPS-Coach

- Designed an offline practice-review pipeline: OBS capture, CUDA YOLO analysis after the session, and a Unity war room that replays the footage with detection overlays and a session-level, **FOV-aware** aim report (crosshair deviation and tracking; deviation at fire, reaction time and overcorrection from audio).
- Validated target selection against **44** server-confirmed headshots on range footage: median error at the shot of **0.36°** when the head is placed from the body box, vs **2.48°** when ranking detected head boxes only.

Simulation-to-real small-object detection 03/2023 – 09/2023

Queen Mary University of London | Supervisors: Dr Jeremy Gow, Dr Haowei Peng

- Unity synthetic-data pipeline (**6000+** annotated scenes) for training a YOLOv5s detector on sub-30 px targets (**93% mAP**), exported through ONNX to Unity Barracuda.

Improved MCTS agent in Pommerman 03/2023 – 06/2023

Queen Mary University of London | Supervisor: Dr Diego Perez-Liebana

Repo: github.com/QizhouD/java-pommerman/tree/master/ImprovedMCTS

- Compared basic AMAF, **α-AMAF** and RAVE enhancements to MCTS in the Java Pommerman framework; α-AMAF raised the win rate by about **5 percentage points** over the benchmark MCTS agent under full observability and was on par under partial observability.

Modeling player navigation in games (critical review) 03/2023 – 06/2023

Queen Mary University of London | Supervisors: Dr Jeremy Gow, Dr Laurissa Tokarchuk

- Surveyed GUR navigation analytics and pathfinding (A*, DPA hybrids, RRT, etc.); case study on The Last of Us NPC AI (NavMesh, exposure maps, sensory models); identified gaps in sim-to-real validation and audiovisual cues.

Dwarf & Fire: C++17 / SFML action game 12/2022 – 03/2023; extended 2025 – 2026

Queen Mary University of London coursework | Instructor: Dr Diego Perez-Liebana

Repo: github.com/QizhouD/2D-action-game

- Component-based entities run through a single fixed-order system pipeline (input, AI, movement, TTL) on a fixed 60 Hz timestep; Command, Observer and Service Locator patterns; unit tests and CI on Windows and Linux.

WORK EXPERIENCE

Software Engineer, China Telecom Ltd., Nanning, China 03/2024 – Present

- Deployed **on-prem LLM + RAG** for context-aware internal data analysis and querying.
- Built a **target recognition** microservices pipeline (deep-learning detection, Flask, Docker, RabbitMQ) for high-throughput video streams.
- Developed Spring Boot / Vue payroll and approval workflows for production systems.

Operations Assistant Intern, China Telecom (Europe) Ltd., London, UK 03/2023 – 05/2023

- Python (Pandas / Matplotlib) analytics and VBA automation for e-commerce reporting.

SKILLS

Programming Languages	C# (Proficient) | Java (Proficient) | Python (Proficient) | C/C++ (Familiar)
Backend Development	Spring Boot | JUnit | RESTful APIs | MySQL | Git | .NET
AI/ML Skills	TensorFlow | PyTorch | Pandas | NumPy | OpenCV | Matplotlib
Frontend Exposure	JavaScript | Vue.js | HTML/CSS
Tools & Platforms	Docker | Linux | Postman | Unity